

UNIVERSITY OF
CAMBRIDGE

MPhil in Economics and Data Science
Module: D100 / D400: Fundamentals of Data Science and Research Computing
Combined Project

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Actual word count (excld. code, plots, tables):

1998

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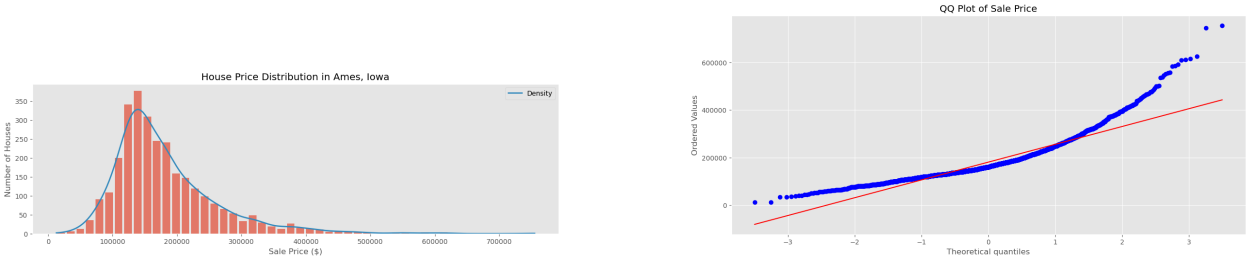
Motivation

The Ames Housing dataset (Decock, 2024) includes 82 features describing 2,930 properties in Ames, Iowa. These encompass property dimensions, facilities, and neighborhood characteristics, offering an excellent opportunity for predictive modeling with heterogeneous data.

Predicting housing prices is useful for real estate development and mortgage lending, but the primary goal is to create an exceptional data science project. Given the strong correlations among features, the challenge lies in feature engineering— transforming individual attributes into meaningful aggregated measures. By deriving new features from the original dataset, this project successfully improves the performance of two competing models: Generalised Linear Models (GLM) and Light Gradient Boosting Machine (LGBM).

Explanatory Data Analysis (EDA)

Target: `saleprice`



(a) Target distribution

(b) Plot 2

Figure 1: QQ Plot

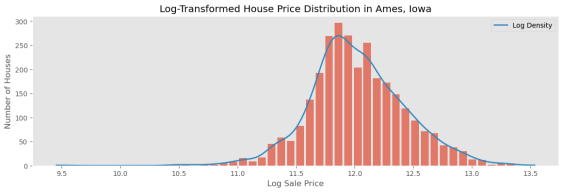
Table 1: Normality Tests ($\alpha = 0.1$)

Test	**Test Statistic**	**p-value**	**Result**
D'Agostino-Pearson	1074.647	4.40×10^{-234}	Reject H_0
Shapiro-Wilk	0.876	2.33×10^{-43}	Reject H_0
Anderson-Darling	87.367	NaN	Reject H_0
Jarque-Bera	4668.421	0.0	Reject H_0

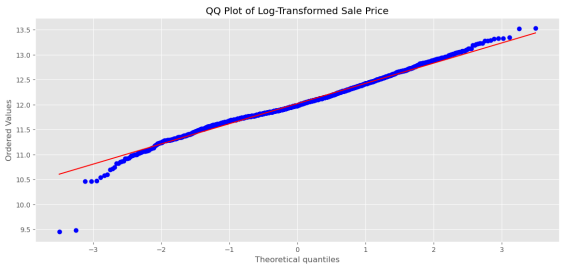
Table 2: Fat-Tailed Tests ($\alpha = 0.1$)

Test	**Test Statistic**	**Reference Value**	**Result**
Kurtosis	8.108	3.0	Fat-tailed

The target variable has no missing values. Approximately 70% of properties have house prices below $200,000, while some exceed $500,000. A series of tests confirm the target's non-normality.



(a) Log-Price Distribution



(b) QQ Plot

Figure 2: Log-Price Analysis

To explore log-normality, we examine the log-transformed prices: The log-transformed distribution remains non-symmetric with long tails. Consequently, a *gamma* distribution with a log-link is selected for the GLM based on the following considerations:

- House prices are strictly positive, which aligns with the domain of the gamma distribution.
- The log-link function flexibly models non-linear, multiplicative relationships between features and the target (Ford 2022). It ensures predictions remain positive while leveraging the gamma distribution's property of increasing variance with the mean. This naturally captures the fact that higher-value properties exhibit proportionally larger variations. For example, prices below \$150,000 are relatively clustered, whereas high-value properties often differ by more than \$200,000.

Cleaning

1) Outliers



Figure 3: Outliers

A plot of sale price against living area reveals five properties with excessively large living areas exceeding 400,000 square feet. Three are partial property sales, while the other two are exceptionally

large houses sold at prices unrepresentative of general market values (Decock, 2024). We will exclude these five observations by filtering for houses with living areas below the threshold.

2) Missing and inconsistent values



Figure 4: Missing values

- The yellow highlights missing values. According to the data dictionary, `NA` represents absent features like pools (90% of houses lack them), alley access, and fences (absent in 80%). Any variables related to these are excluded to improve the models' generalisability.
- For missing `bsmt` (basement) variables, areas are recorded as zero. Similarly, `mas_vnr` (masonry veneer) and `fireplace_qu` (fireplace quality) are missing when the number of fireplaces is zero. These were imputed using a `simple_keywords_imputer`, assigning `NONE` for categorical and `0` for numerical variables.
- One missing `electrical` is imputed with `SBrkr` matching others in the same neighborhood and building class.
- `garage_year_blt` is filled with `year_blt`. Observation 2260's `garage_year_blt` is corrected from 2207 to 2007 due to a likely typo.
- Filtered out observations where `garage_year_blt` or `year_blt` abnormally exceed `year_sold`.
- `garage_type` misses exactly where `garage_area` is zero, so filled with `NONE`.

- Figure 5 shows that `lot_frontage`, typically determined by neighborhood, exhibits minimal variation within neighborhoods and configurations. We implement a `LotFrontageGroupMedianImputer` to impute values based on the conditional median of the two variables, applied after data splitting to prevent leakage.

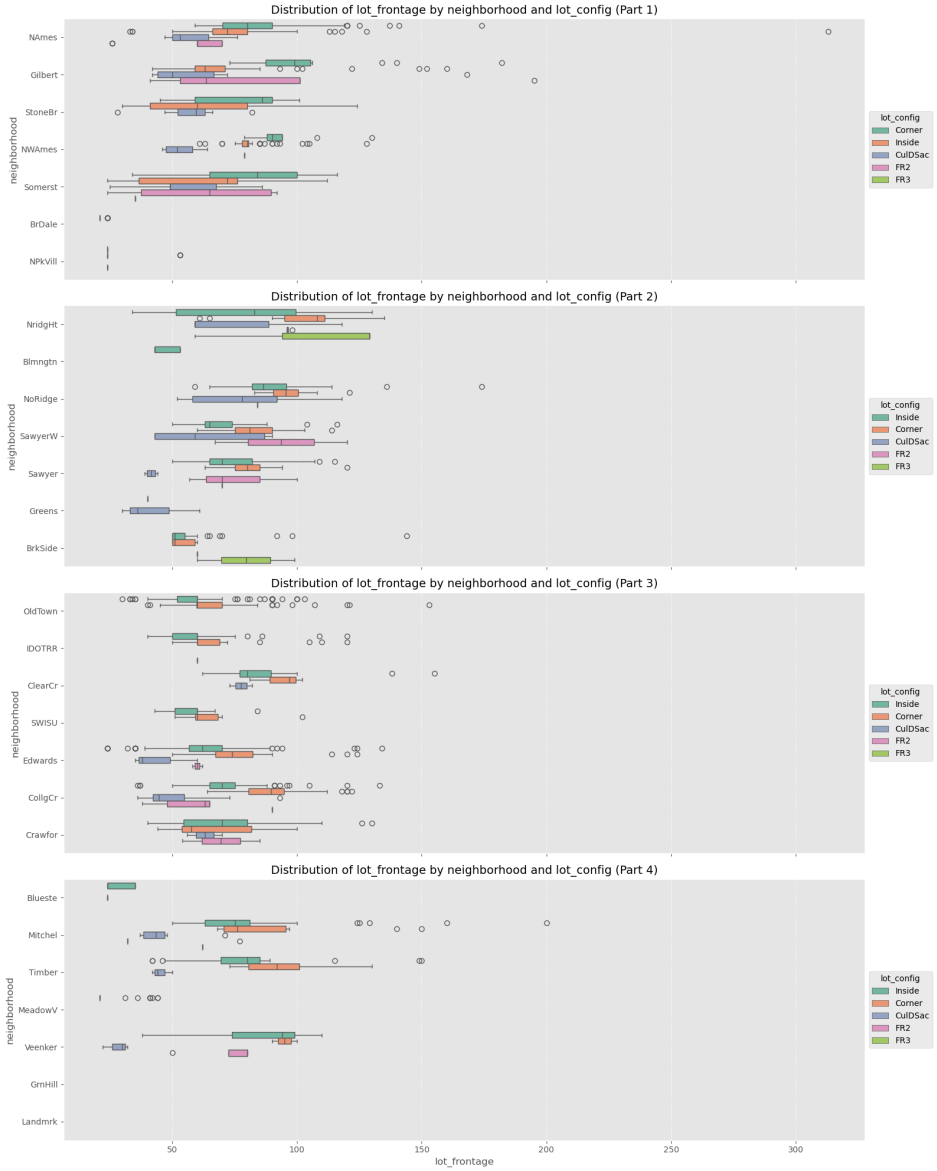


Figure 5: Lot frontage by neighbourhood and configurations

Feature Selection

1) Distribution

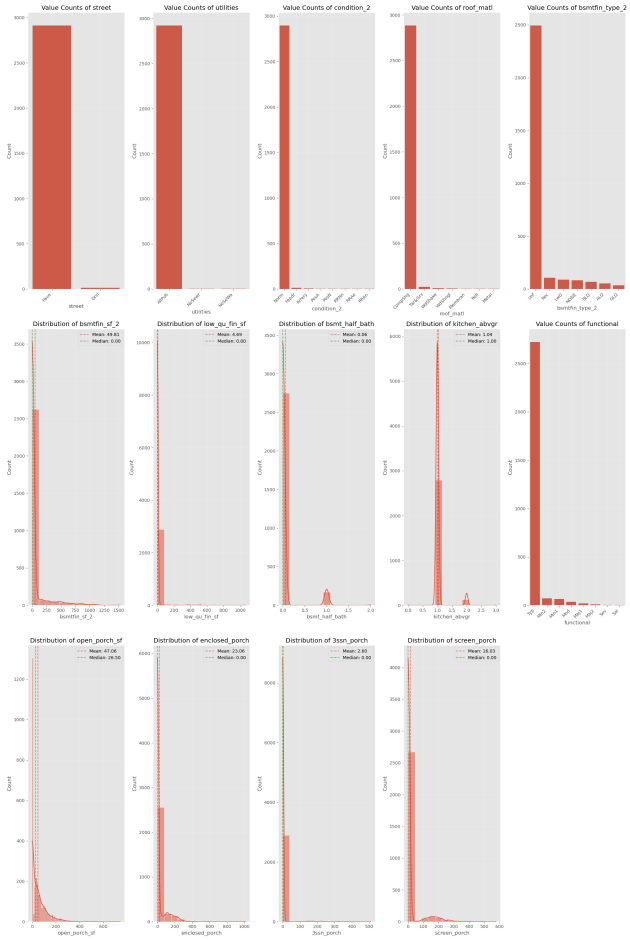


Figure 6: Features with few variations

- Remove categorical features with nearly uniform counts, lacking discriminatory power.
- Drop count features where most houses lack the attribute (e.g., porches, half baths in basements, or multiple kitchens).

2) Categorical features

ANOVA

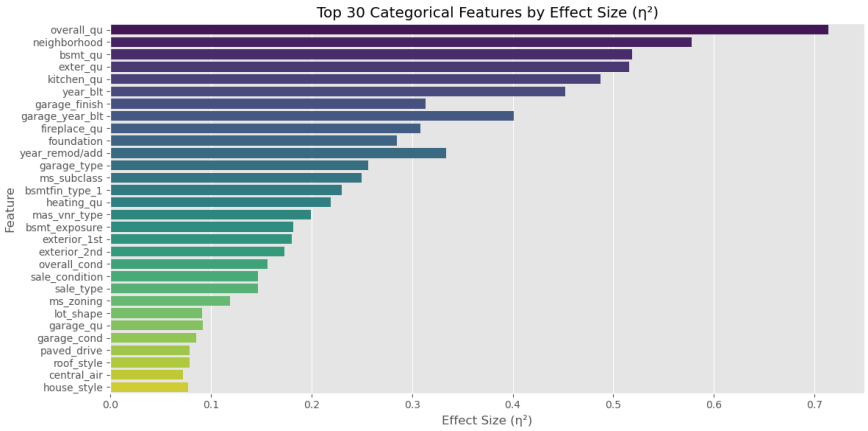


Figure 7: One-way ANOVA

This assesses variance differences between groups of a categorical feature for a continuous target, providing η^2 to measure dispersion. Powerful features would have differing variations. The method assumes normality within groups, which may not hold. However, the results align with domain knowledge. For example, `central_air` (air conditioner) likely havs minimal impact on housing price because it can be easily added.

Cramér's V

This uses a chi-squared statistic capturing how much two categorical variables deviate from independence. The result aligns with domain knowledge: `bldg_type` and `ms_subclass` both indicate building structures. Similarly, despite imputing only 153 missing `garage_year_blt` values with `year_blt`, strong correlation reflects garages typically being built with houses.

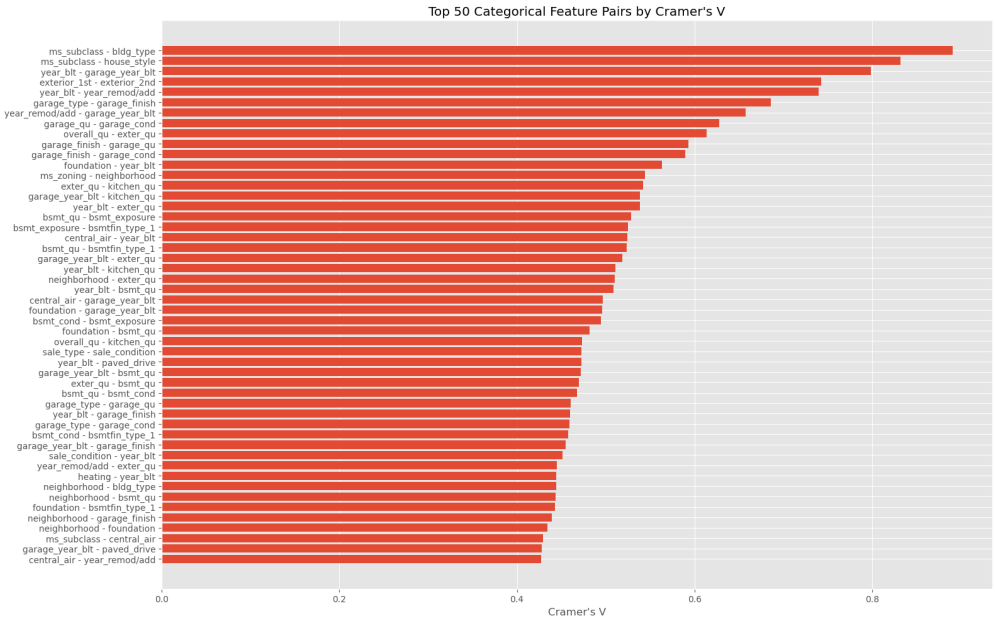


Figure 8: Features with few variations

`ms_zoning` (residential zone classification) largely overlaps with neighbourhood, mostly being `RL` (low density), while other categories are rare. We drop this redundancy.

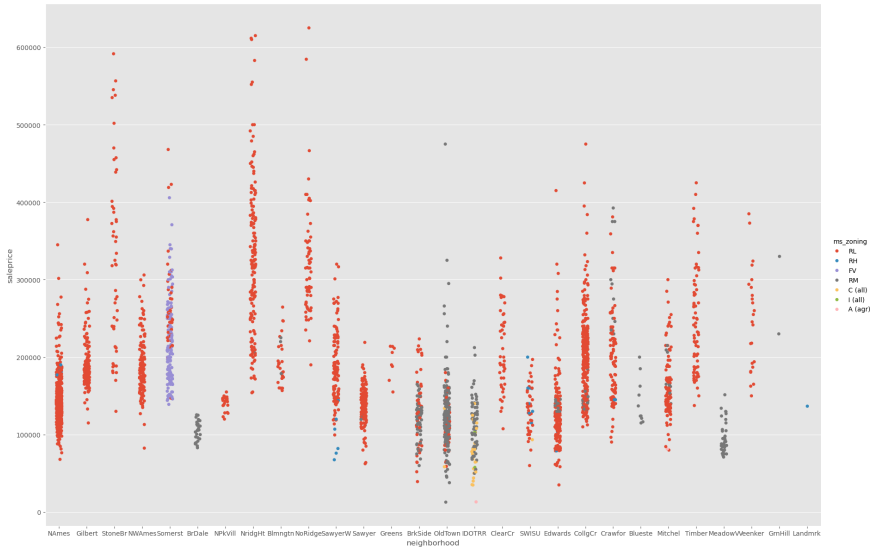


Figure 9: Residential Zones by Neighbourhood

3) Numerical features

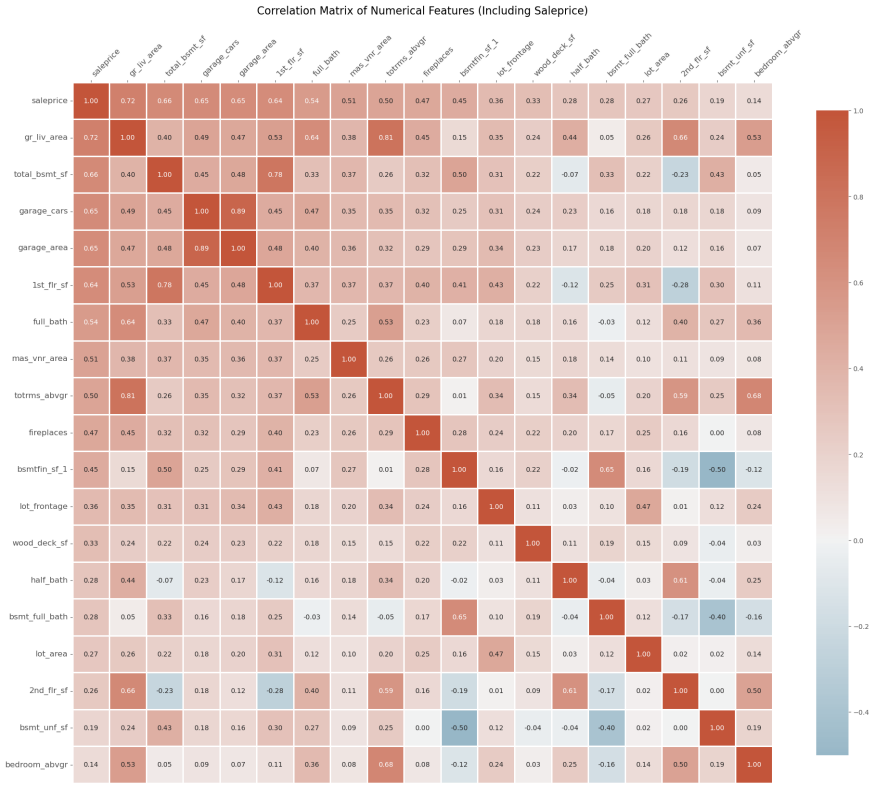


Figure 10: Correlation Matrix of Numerical Features

Features strongly correlated with `saleprice` are also highly intercorrelated. Redundancies include `garage_car` and `garage_area`, `1st_flr_sf` and `total_bsmt_sf` (as basements are directly below), `totrms_abv_gr` (total rooms) and gross living area, and finished vs. unfinished basement area (`bsmt_unf_sf`).

By exploring these redundancy, we managed to arrive at a final set of most relevant features:

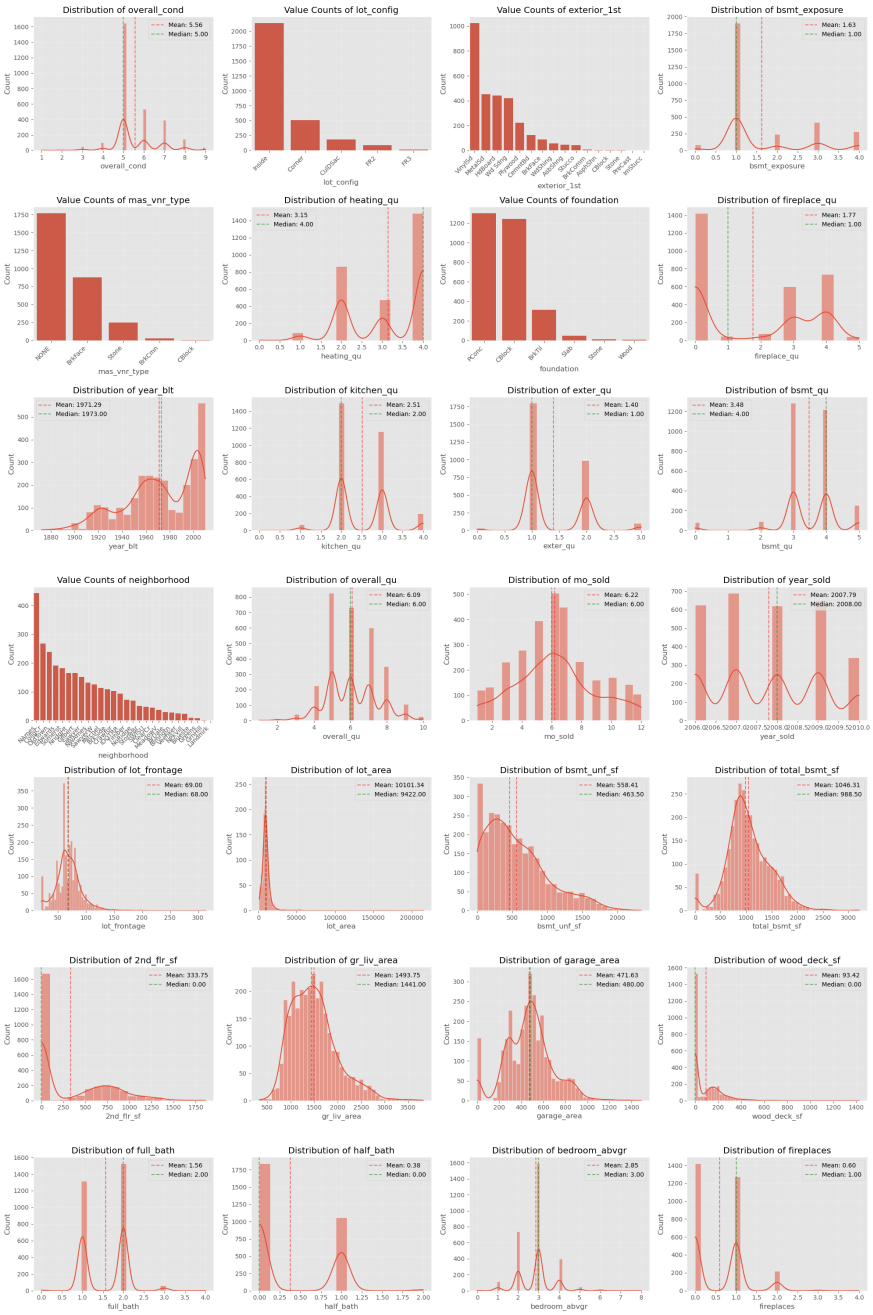


Figure 11: Distribution of final sets of features

In the above, ordinal quality categories (`_qu`) and `bsmt_exposure` (garden access from the basement) are re-recoded to a standardized numeric scale using the dictionary and the custom function `type_formatting()`.

Feature engineering

- From Figure 11, very few houses have half baths. We create `total_bath = full_bath + half_bath` to capture total bathroom usage without redundancy.
- Aggregate `overall_cond` and `overall_qu` into `overall_score` to represent house quality and condition collectively, reducing redundancy while capturing overall desirability.
- Replace 28 neighborhood categories with two new features:
  - `university_proximity_category`: Indicates proximity to the University of Iowa, reflecting appeal to younger populations while accounting for reduced desirability among high-paying buyers.
  - `neighbourhood_score`: Uses the mean `overall_score` for each neighborhood, representing desirability as an ordinal scale instead of multiple dummy variables.
- Create `interior_qu` by aggregating features like `kitchen_qu`, `heating_qu`, `fireplace_qu`, `bsmt_qu`, and `bsmt_exposure`, improving interpretability while capturing the strong correlations between interior conditions.
- Define `age = year_sold - year_blt`, as a house's age at purchase is more relevant to price than its construction year.
- Create `pct_unf_sf = bsmt_unf_sf / total_bsmt_sf` to quantify the basement's level of finishing.
- Transform `woodeck_sf` and `2nd_flr_sf` into dummies (`has_woodeck` and `has_2nd_flr`), as very few houses have these features.
- Aggregate months sold into seasonal indicators to capture systematic price patterns, with prices typically declining in winter and rising in summer.

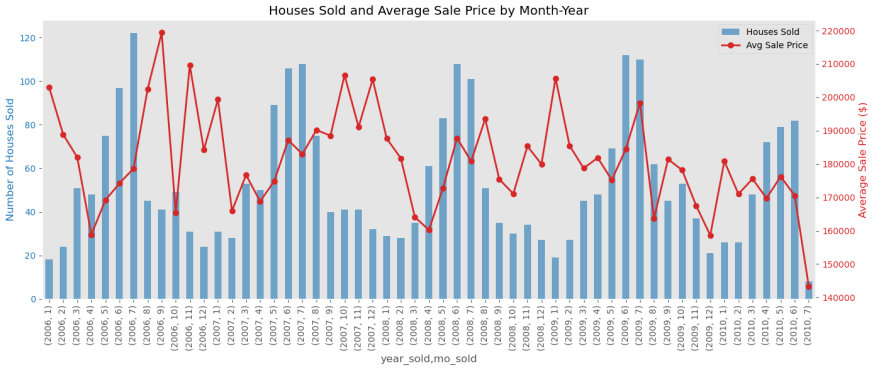


Figure 12: Sale Price by Month-Year

As `neighbourhood_score` compares across observations, to avoid data leakage, we provide `Add_Drop_Attributes` transformer that easily included in the pipeline to aggregate features.

1 GLMs

We start by imputing lot frontage and adding categorical dummies and train a baseline on 28 originally selected features.

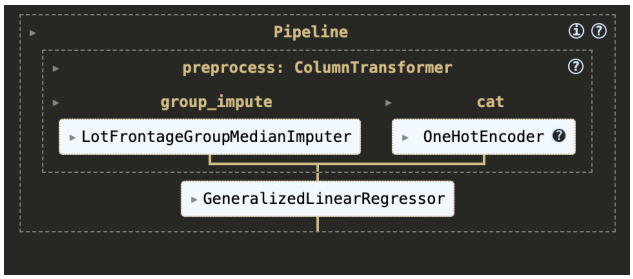


Figure 13: Baseline GLM pipeline

We then aggregate features and use a custom `WinsorizedRobustScaler` to clip extreme outliers (1st and 99th percentiles), center by the median, and scale by the interquartile range. This mitigates the impact of extremes before ensuring robustness, as moderate outliers will not significantly affect the scaling.

For ordinal variables, we created a custom `OrdinalMerger` transformer that dynamically aggregates infrequent categories into the next higher one, preserving the ranking.

For nominal variables, we group infrequent categories as `Other` and apply one-hot encoding.

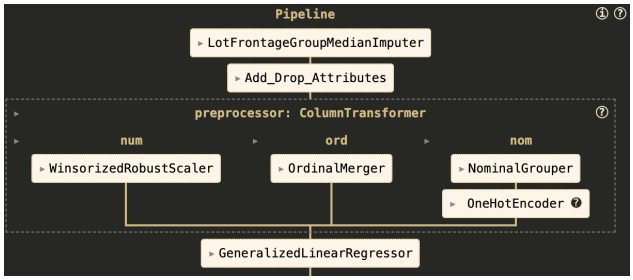


Figure 14: GLM pipeline with transformers

While GLMs are flexible, we may improve the model by pre-defining some interesting interactions.

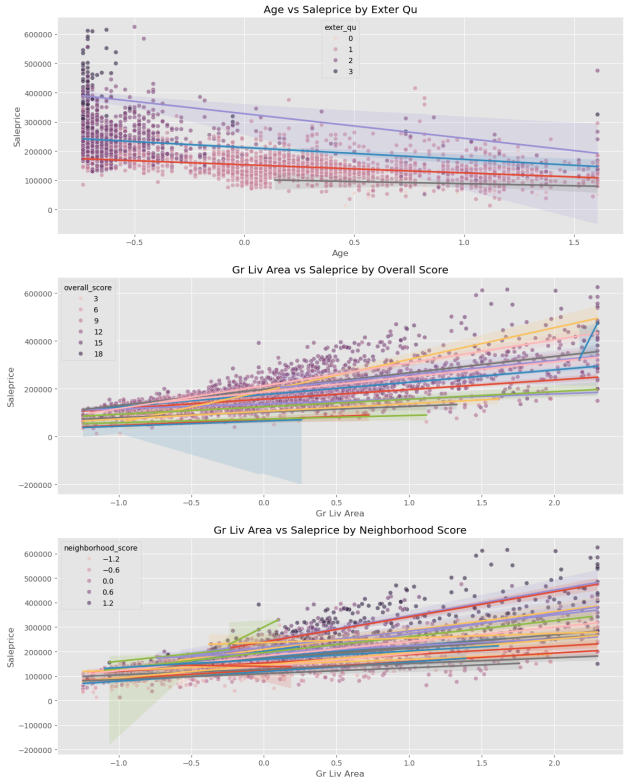


Figure 15: Interactions

Interestingly, older houses with expensive exterior materials depreciate faster with age. Price rises more sharply with larger living areas, the higher overall quality, and greater neighborhood prestige.

We include interactions and the final model is fine-tuned by cross-validation using `GeneralizedLinearRegressorCV(cv=5)` to get the optimal regularisation `alpha=0` and `l1_ratio=0.0001`.

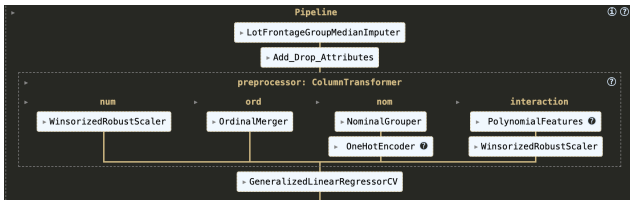


Figure 16: Final GLM pipeline

Model	RMSE	RMSED (%)	MAPE (%)	MedAE	R-squared	Bias	Deviance
Baseline	39739.53	21.92%	16.47%	19772.31	0.75	-109.70	1.58e+09
Agg.Features	21330.47	11.77%	9.21%	10413.38	0.93	80.56	4.55e+08
Agg.Features w. Inter	20658.86	11.40%	8.86%	10189.05	0.93	-23.69	4.27e+08
Agg.Features w. Inter(tuned)	20778.69	11.46%	8.92%	10152.60	0.93	-58.89	4.32e+08

Table 3: Performance metrics for GLMs on train set

Model	RMSE	RMSED (%)	MAPE (%)	MedAE	R-squared	Bias	Deviance
Baseline	44376.30	25.23%	16.92%	18936.76	0.64	-4433.45	1.97e+09
Agg.Features	22821.78	12.98%	9.37%	10450.47	0.90	307.96	5.21e+08
Agg.Features w. Inter	22595.60	12.85%	9.06%	10250.87	0.91	91.89	5.11e+08
Agg.Features w. Inter(tuned)	22488.98	12.79%	8.94%	9995.30	0.91	23.45	5.06e+08

Table 4: Performance metrics for GLMs on test set

The second and third metrics are scaled RMSE and MedAE. Moving from the baseline to aggregated features reduces all metrics, demonstrating the strategy's success. Adding interactions further improves performance, while tuning regularization reduces overfitting and enhances test performance, making it the final GLM.

LGBMs

LGBMs allows for minimal processing, so the baseline is

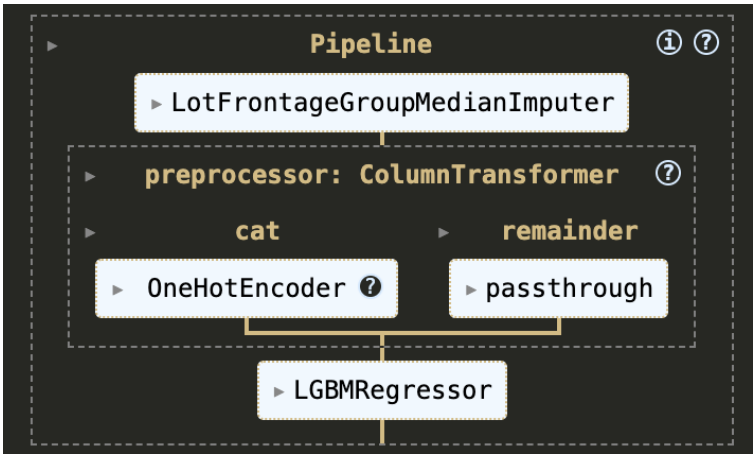


Figure 17: Baseline LGBM pipeline

and one using aggregated features, further tuned `learning_rate, n_estimators, n_leaves and min_child_weight` by `optuna`- the objective uses RMSE with K-Fold CV.

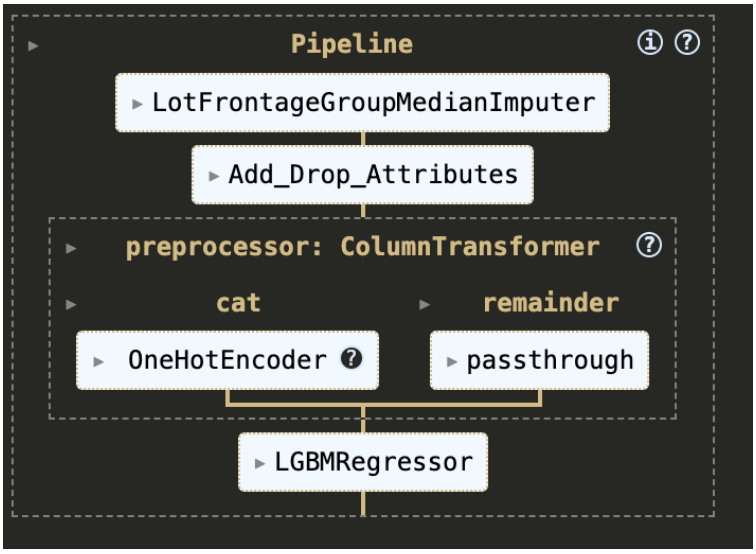


Figure 18: LGBM with aggregated features pipeline

We further experimented with a custom objective loss for LGBM (Yan [2023], Yan [2023]), replacing `gamma`:

```
def custom_regression_loss(y_true, y_pred):
    # Calculate the residual (error)
    residual = y_pred - y_true
    # Calculate the gradient (first-order derivative)
    gradient = 2 * residual
    # Calculate the hessian (second-order derivative)
    hessian = 2 * np.ones_like(y_true)
    # Define a simple linear penalty term
    penalty = np.abs(residual) * 0.05
    # Combine the gradient and penalty
    gradient += penalty
    return gradient, hessian
```

This resembles quadratic loss but adds a linear penalty to residuals, making the model more sensitive to handle outliers. It is further optimised using RandomizedCV. The baseline model exhibits significant bias, while the tuned LGBM with `gamma` objective and aggregated features achieves lower bias and comparable deviance. The custom objective performs better than expected but falls short of the `gamma` objective. The strong performance of the baseline highlights LGBM's ability to handle correlated features effectively. We will proceed with the tuned LGBM using the `gamma` objective for consistency with the features used in the GLM.

Model	RMSE	RMSED (%)	MAPE (%)	MedAE	R-squared	Bias	Deviance
Baseline LGB	21641.58	12.30%	8.55%	9408.22	0.91	-1359.24	4.69e+08
tuned.agg LGB	22101.73	12.57%	9.00%	10416.65	0.91	1144.28	4.88e+08
tuned.agg LGB custom	22714.23	12.91%	9.49%	11252.90	0.90	1369.86	5.16e+08

Table 5: Performance metrics for LGBMs on test set

Evaluation

Performance

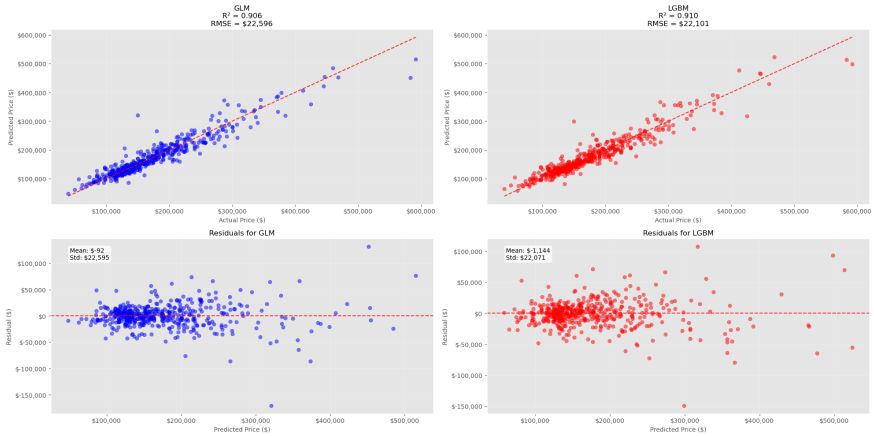
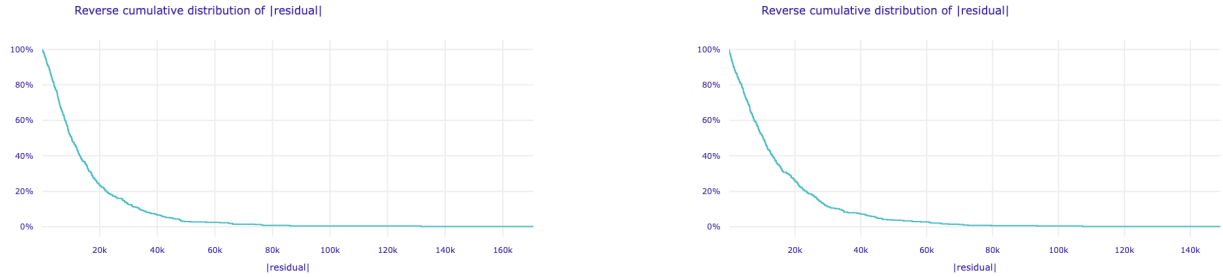


Figure 19: Predicted v. Actual

LGBM slightly outperforms GLM in tracking actual values on the test set, with predictions overestimating by up to $100,000 and underestimating by around $150,000. However, both residual plots show increased variance in the higher price range, indicating heteroskedasticity not adequately addressed by the current pipelines. This may stem from over-filtering rare features potentially explaining variations among higher-priced houses.



(a) GLM

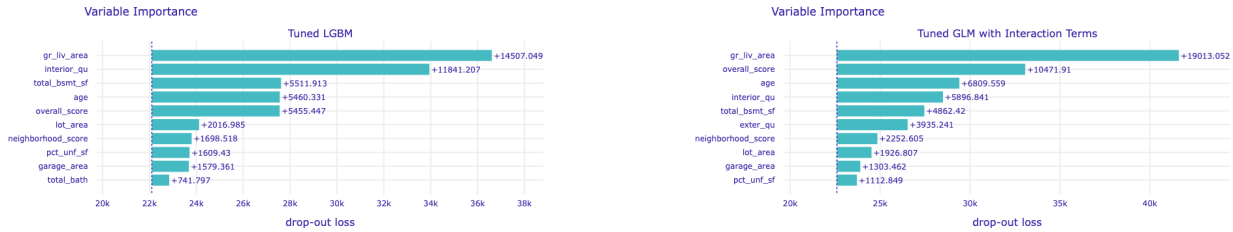
(b) LGBM

Figure 20: Reverse cumulative distribution of residuals (on train set)

This shows the percentage of predictions with residuals equal or larger in absolute values to a

specific value on `x-axis`. LGBM gives more fraction of residuals before $20,000 and the absolute magnitude that residuals can take is less than GLM. We therefore select LGBM.

Feature Importance



(a) Feature Importance - LGBM

(b) Feature Importance - GLM

Figure 21: Comparison of Feature Importance for LGBM and GLM Models

Both models agree on the top five most important features, whose absence would significantly impair performance (measured in drop-out loss): `gr_liv_area`, `interior_qu`, `total_bsmt_sf`, `age`, and `overall_score`. As expected, these features—representing living and basement area, building age, and quality scores—are highly relevant in predicting a property's price.

Partial Dependence Plots (PDPs) of Top Features

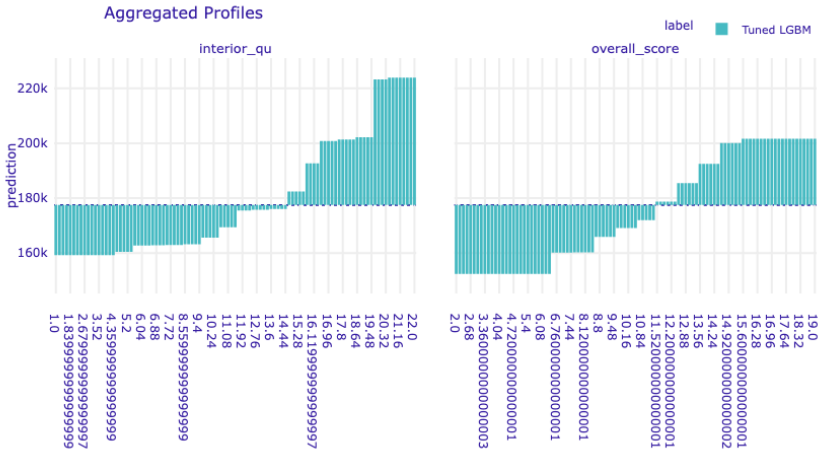
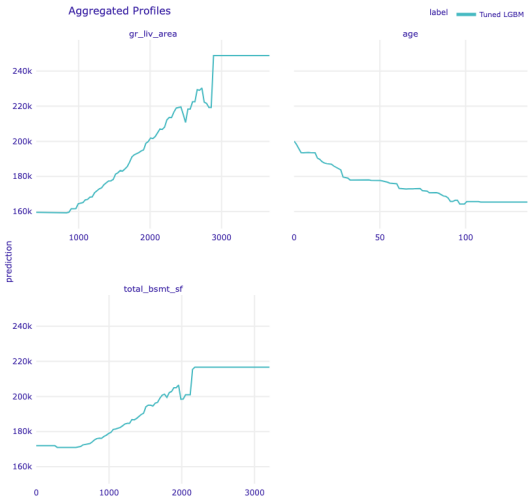


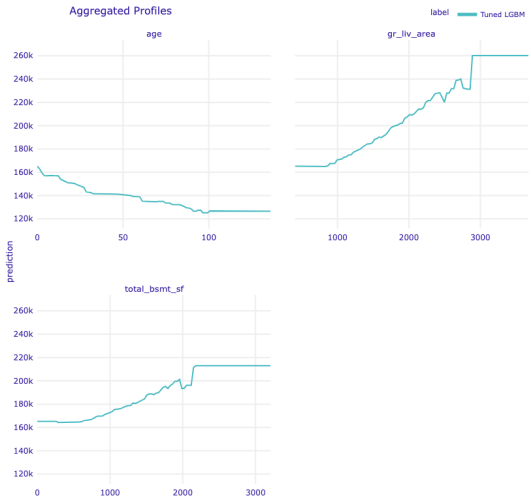
Figure 22: LGBM PDP Ordinal Features

This can be interpreted as the expected average response of price as a function of features, marginalizing over the values of all other top ones (Molnar, 2019). Without explicitly enforcing a monotonicity constraint, the model successfully captures the intuitive relationship that higher quality

scores tend to drive prices upward.



(a) LGBM PDP Continuous Features

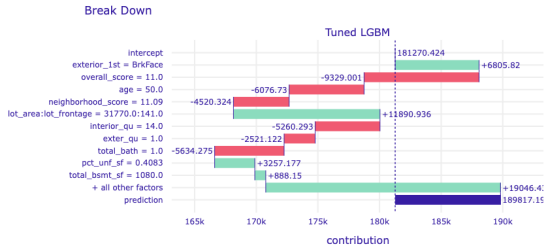


(b) LGBM ADP Continuous Features

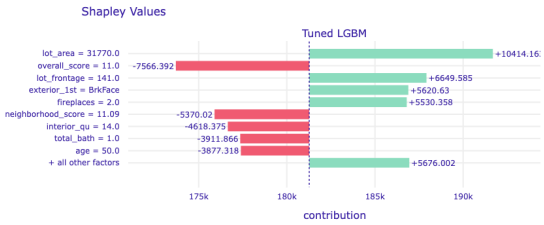
Figure 23: Comparison of PDP and ADP

Price increases with house area (basement and gross) but decreases with age, reflecting depreciation. The effect stabilises for very large or very old houses, where further changes have little impact on price. Feature correlations likely violate PDP assumptions, introducing bias, though ADPs address this and show similar patterns. Adding a monotonicity constraint on house area could further enhance the model, as larger houses are expected to cost more and we would have smoother PDPs.

Contribution



(a) LGBM Break Down



(b) LGBM Shapley values

Both methods explain model predictions by attributing the final output to feature contributions for a single observation. Break Down sequentially adds or subtracts each feature's contribution starting from a baseline (intercept). Shapley values, a generalization, distribute the difference between the prediction and baseline across all features, considering all combinations (Kozak, 2024). Features like large lot area, ample frontage, and two fireplaces increase value, reflecting good facilities. In

contrast, an average overall score, advanced age (50 years), and low interior quality reduce value, consistent with expectation that depreciated conditions make it lose appeal.

Finally, the CP profiles in Figure 25 show the model's response at a single point in the feature space, highlighting how predictions vary with changes in one input variable while others remain constant. Again, features such as age, overall condition (interior and exterior), neighborhood, and area have the largest impact. Notably, the profiles confirm a response to the seasonal indicator for summer months, as expected from earlier discussions as the peak time for transaction.

Improvements

The models effectively predict house prices, with aggregated features enhancing interpretability and providing clear insights into key influence of area, age, condition, and neighborhood. They can be improved by expanding the range of tuning parameters and incorporating more features with sufficient variation among high-end house prices to improve prediction there. Addressing the residual heteroskedasticity may also require less stringent feature filtering and better engineering of rare features into an index that highlights their impact on property value while mitigating collinearity. Additionally, techniques such as mixed-PCA for heterogeneous data types, or tree-based feature selection could be explored. The strong performance of the custom loss function for LGBM suggests further experimentation, including the use of Huber loss, which is less sensitive to outliers and reduces informational loss from winsorising.

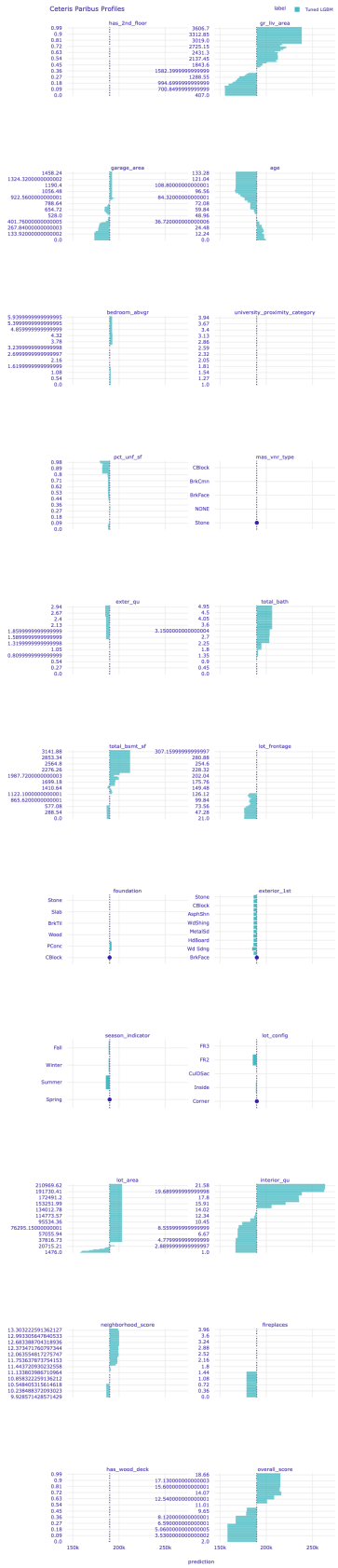


Figure 25: Ceteris Paribus Profiles

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